

TASK 02

BRAINWAVE MATRIX SOLUTIONS

CLOUD COMPUTING INTERN

11- Cloud Computing Intern

E-commerce site or any static website with cloud storage

This project involves creating a simple HTML/CSS website and deploying it to Google Cloud.

OR

Chat Application with Firebase

Create a real-time chat application using Firebase. This project involves using Firebase services like Firestore, Authentication, and Hosting.

Report submission in 15-20 days.

- Go to the AWS console management dashboard.
- Select **S3** from there.
- Create an **S3 bucket**.
- Give bucket's name as "**shreya-ecommerce-site**".
- Uncheck the "block public access for this site".
- Acknowledge the warning.
- Click on create.

The image shows two screenshots of the AWS Management Console. The top screenshot displays the 'S3 buckets' page for the 'us-east-2' region. It includes a sidebar with navigation options like 'General purpose buckets', 'Directory buckets', and 'Storage Lens'. The main content area shows an 'Account snapshot' and a list of 'General purpose buckets'. One bucket is listed: 'elasticbeanstalk-us-east-1-413240597866' in the 'US East (N. Virginia) us-east-1' region, created on 'October 2, 2023, 19:57:30 (UTC+05:30)'. The bottom screenshot shows the 'Create bucket' wizard. Under 'General configuration', the 'AWS Region' is 'US East (Ohio) us-east-2'. The 'Bucket type' is set to 'General purpose'. The 'Bucket name' is 'shreya-e-commerce-site'. The wizard also includes a section for 'Copy settings from existing bucket - optional'.

Amazon S3

General purpose buckets
Directory buckets
Table buckets
Access Grants
Access Points
Object Lambda Access Points
Multi-Region Access Points
Batch Operations
IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens
Dashboards

Account snapshot - updated every 24 hours All AWS Regions [View Storage Lens dashboard](#)

Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets | Directory buckets

General purpose buckets (1) Info All AWS Regions [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	IAM Access Analyzer	Creation date
elasticbeanstalk-us-east-1-413240597866	US East (N. Virginia) us-east-1	View analyzer for us-east-1	October 2, 2023, 19:57:30 (UTC+05:30)

Create bucket Info

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (Ohio) us-east-2

Bucket type Info

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name Info
shreya-e-commerce-site

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

us-east-2.console.aws.amazon.com/s3/bucket/create?region=us-east-2&bucketType=general

Amazon S3 > Buckets > Create bucket

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.

☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Turning off block all public access might result in this bucket and the objects within becoming public
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☐ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

☒ **Server-side encryption with Amazon S3 managed keys (SSE-S3)**
☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the [Storage](#) tab of the [Amazon S3 pricing page](#).

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable
☒ Enable

Advanced settings

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel Create bucket

- Go inside the bucket.
- Click on **upload**, add files.
- Upload the html file(**index.html**) in the bucket.

The screenshot shows the AWS Management Console interface. At the top, a green banner indicates the successful creation of the bucket "shreya-e-commerce-site". Below this, the "General purpose buckets" section is active, displaying a table of buckets. The bucket "shreya-e-commerce-site" is highlighted. Below the table, the details for the "shreya-e-commerce-site" bucket are shown, including tabs for Objects, Metadata, Properties, Permissions, Metrics, Management, and Access Points. The "Objects" tab is selected, showing an empty list of objects with an "Upload" button.

General purpose buckets (1/2) [Info](#) [All AWS Regions](#)

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	IAM Access Analyzer	Creation date
elasticbeanstalk-us-east-1-413240597866	US East (N. Virginia) us-east-1	View analyzer for us-east-1	October 2, 2023, 19:57:30 (UTC+05:30)
shreya-e-commerce-site	US East (Ohio) us-east-2	View analyzer for us-east-2	April 14, 2025, 19:54:54 (UTC+05:30)

shreya-e-commerce-site [Info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [Access Points](#)

Objects (0) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
No objects				

You don't have any objects in this bucket.

[Upload](#)

The screenshot displays the AWS S3 console's 'Upload' page for the bucket 'shreya-ecommerce-site' in the 'us-east-2' region. The interface includes a breadcrumb trail (Amazon S3 > Buckets > shreya-ecommerce-site > Upload), a search bar, and a navigation bar with various AWS services. The main content area is titled 'Upload' and provides instructions on how to upload files. A dashed box indicates where to drag and drop files. Below this, the 'Files and folders (0)' section shows a table with columns for Name, Folder, Type, and Size. A message states 'No files or folders' and 'You have not chosen any files or folders to upload.' The 'Destination' section shows the bucket path 's3://shreya-ecommerce-site'. A second screenshot below shows the same interface after uploading a file named 'index.html' (2.2 KB, text/html type).

Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (0) Remove Add files Add folder

All files and folders in this table will be uploaded.

Find by name

☐	Name	Folder	Type	Size
No files or folders				
You have not chosen any files or folders to upload.				

Destination Info

CloudShell Feedback

Type here to search

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25°C Haze 19:56 14-04-2025

Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (1 total, 2.2 KB) Remove Add files Add folder

All files and folders in this table will be uploaded.

Find by name

☐	Name	Folder	Type	Size
☐	index.html	-	text/html	2.2 KB

Destination Info

Destination

s3://shreya-ecommerce-site

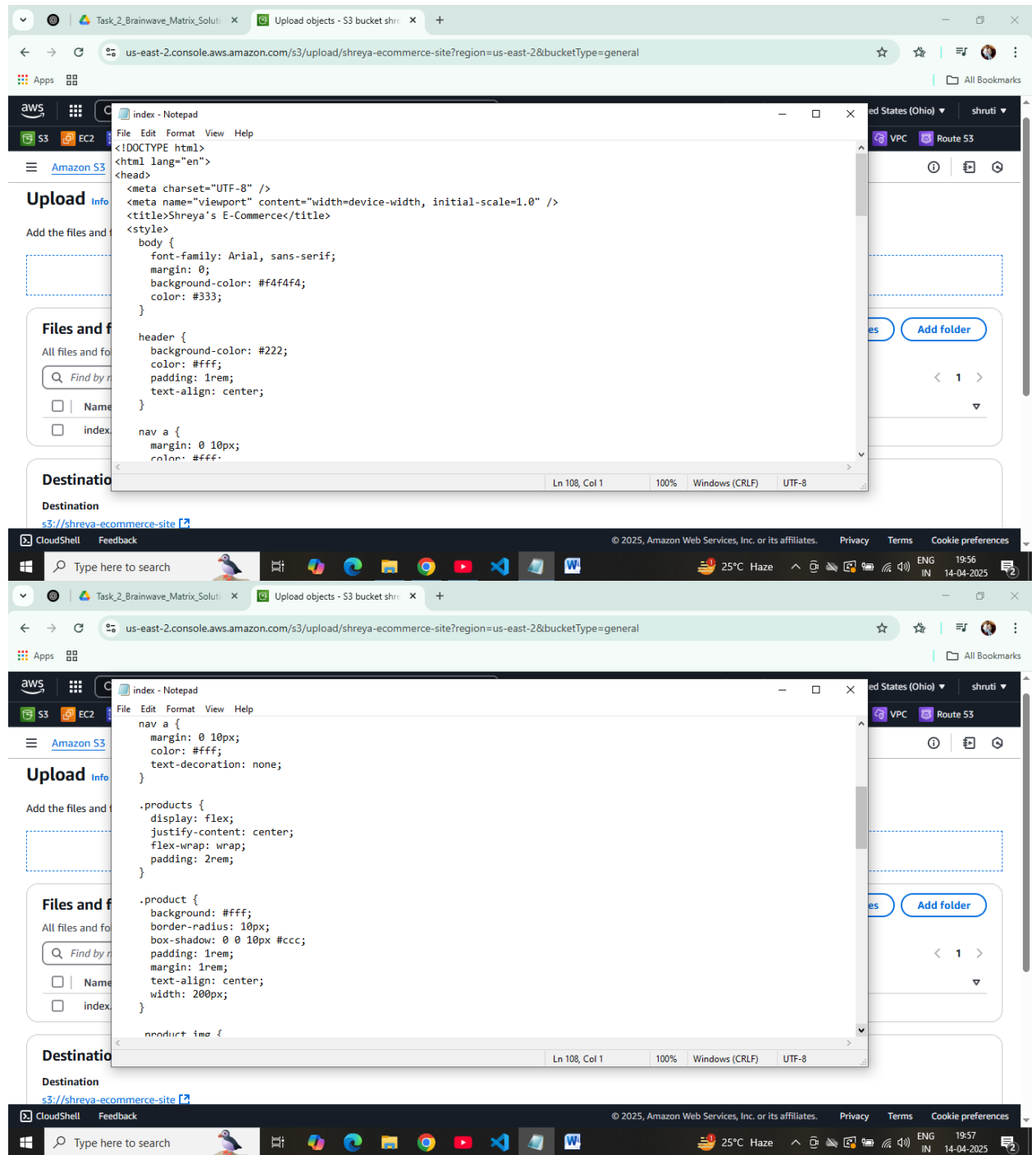
CloudShell Feedback

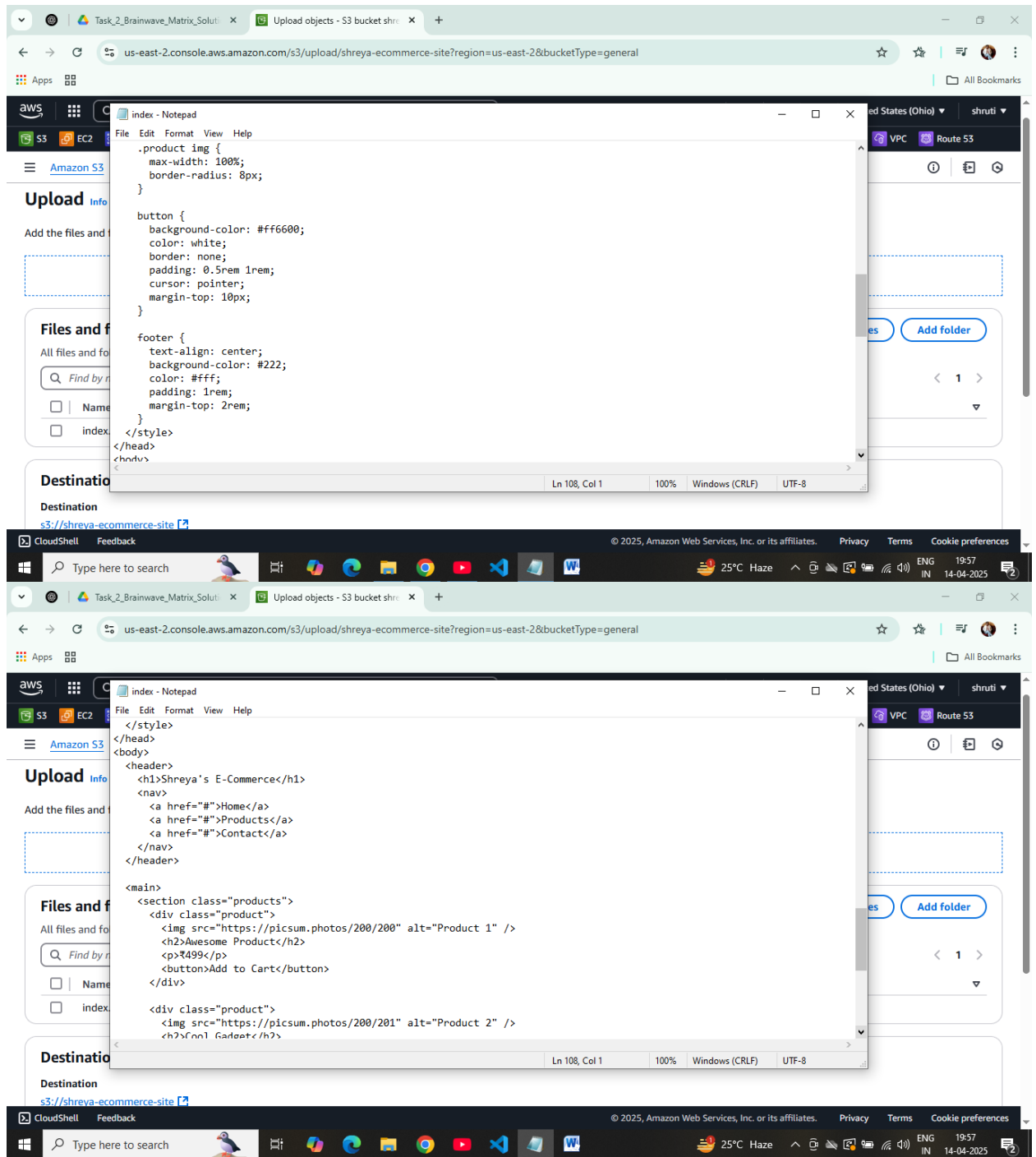
Type here to search

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25°C Haze 19:56 14-04-2025

Shreya Sahay





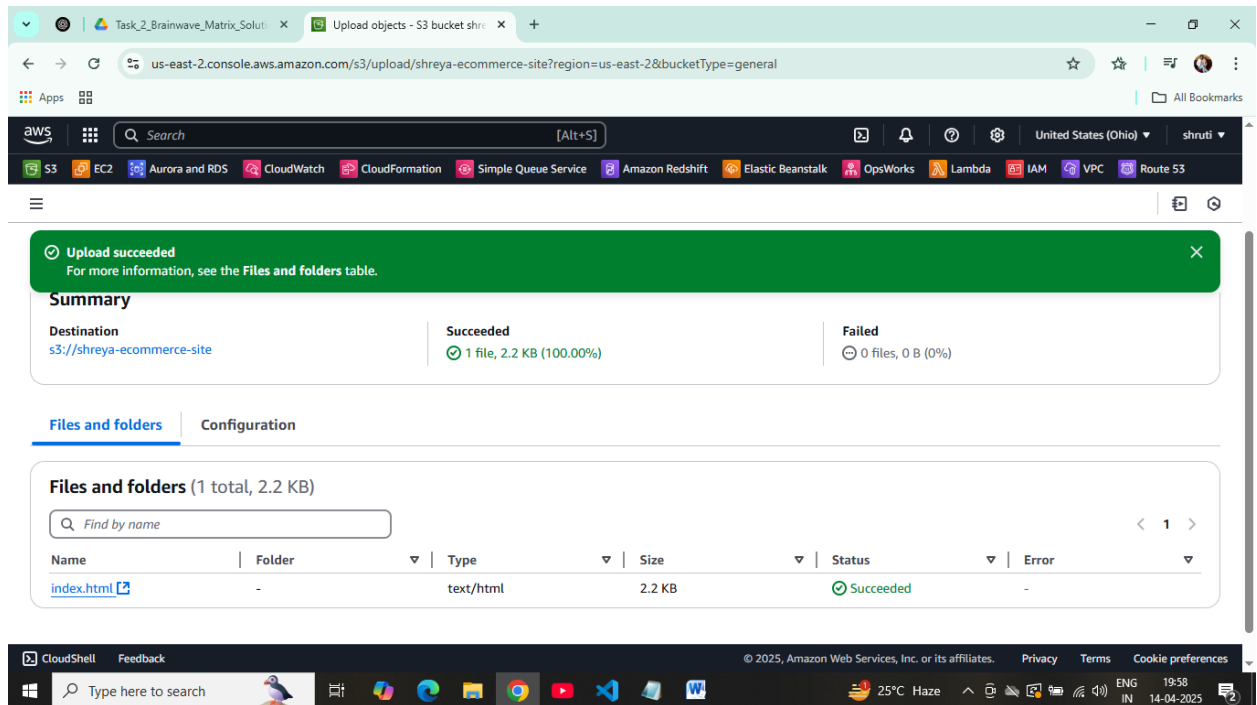
The screenshot displays the AWS S3 console interface for uploading files to a bucket named 'shreya-ecommerce-site'. A Notepad window is open in the foreground, showing the following HTML code:

```
<div class="product">
  
  <h2>Cool Gadget</h2>
  <p>$799</p>
  <button>Add to Cart</button>
</div>

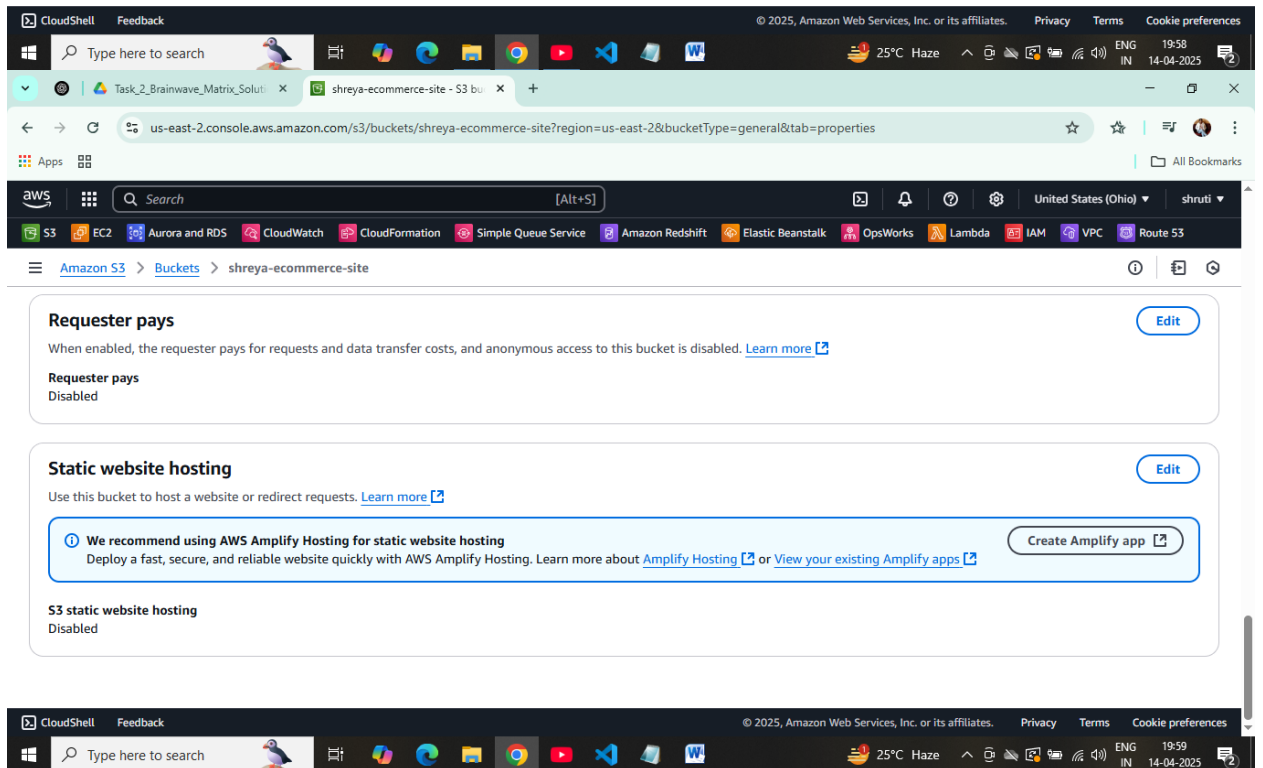
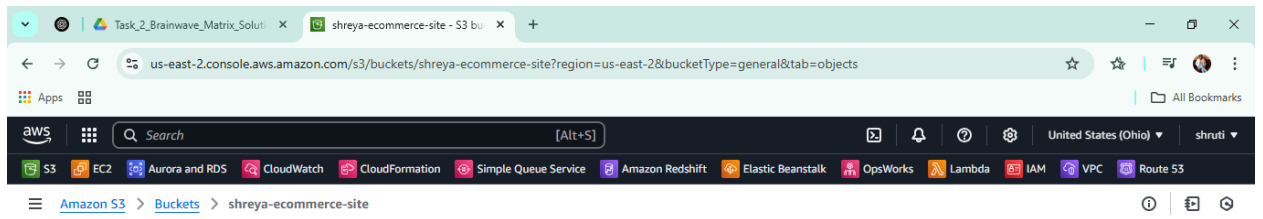
<div class="product">
  
  <h2>Useful Item</h2>
  <p>$299</p>
  <button>Add to Cart</button>
</div>
</section>
</main>

<footer>
  <p>© 2025 Shreya's E-Shop. All rights reserved.</p>
</footer>
</body>
</html>
```

The background shows the AWS S3 console with the 'Upload' tab selected. The 'Destination' section indicates the upload path is 's3://shreya-ecommerce-site'. The 'Permissions' and 'Properties' sections are visible but not expanded. The 'Upload' button is highlighted in orange.



- The file has been uploaded.
- Now, go to the bucket, click on **properties**.
- Go to **static website hosting**, click on edit.
- **Enable** static website hosting.
- Give index document as **index.html** and error document as **error.html**.
- Go to the bucket, click on **permissions**.
- Go to **object ownership**, click on edit.
- Select on **ACLs enabled**.
- Acknowledge that ACLs will be restored.
- Click on save changes.



The screenshot displays the AWS Management Console interface for editing static website hosting on an S3 bucket. The browser address bar shows the URL: `us-east-2.console.aws.amazon.com/s3/bucket/shreya-e-commerce-site/property/website/edit?region=us-east-2&bucketType=general`. The console header includes the AWS logo, a search bar, and a navigation menu with services like S3, EC2, Aurora and RDS, CloudWatch, CloudFormation, Simple Queue Service, Amazon Redshift, Elastic Beanstalk, OpsWorks, Lambda, IAM, VPC, and Route 53. The breadcrumb trail indicates the path: `Amazon S3 > Buckets > shreya-e-commerce-site > Edit static website hosting`.

The main content area is titled "Edit static website hosting" with an "Info" link. It contains the following sections:

- Static website hosting**: A description stating "Use this bucket to host a website or redirect requests." with a "Learn more" link. Below this are two radio buttons: "Disable" and "Enable" (which is selected).
- Hosting type**: Two radio buttons: "Host a static website" (selected) and "Redirect requests for an object". The "Host a static website" option includes a sub-description: "Use the bucket endpoint as the web address. Learn more".
- Informational message**: A blue-bordered box with an information icon stating: "For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see Using Amazon S3 Block Public Access".
- Index document**: A section titled "Index document" with a sub-description "Specify the home or default page of the website." and a text input field containing "index.html".
- Error document - optional**: A section titled "Error document - optional" with a sub-description "This is returned when an error occurs." and a text input field containing "error.html".
- Redirection rules - optional**: A section titled "Redirection rules - optional" with a sub-description "Redirection rules, written in JSON, automatically redirect webpage requests for specific content. Learn more".

The bottom of the screenshot shows a Windows taskbar with various application icons, a search bar, and system tray information including the date (14-04-2025) and time (19:59).

The screenshot displays the AWS Management Console interface for the 'shreya-ecommerce-site' S3 bucket. The 'Permissions' tab is selected, showing the 'Permissions overview' section. Below this, the 'Block public access (bucket settings)' section is visible, with the 'Block all public access' toggle set to 'Off'. The 'Object Ownership' section shows 'Bucket owner enforced' as the selected setting. The 'Access control list (ACL)' section is also visible, with a note indicating that the bucket owner enforced setting is applied for Object Ownership. The interface includes a top navigation bar with the AWS logo, a search bar, and various service icons. The bottom of the screen shows a Windows taskbar with several open applications and a system tray displaying the date and time.

us-east-2.console.aws.amazon.com/s3/buckets/shreya-ecommerce-site?region=us-east-2&bucketType=general&tab=permissions

shreya-ecommerce-site

Permissions overview

Access finding
Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#).
[View analyzer for us-east-2](#)

Block public access (bucket settings) [Edit](#)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access
Off

Object Ownership [Info](#) [Edit](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership
Bucket owner enforced
ACLs are disabled. All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

Access control list (ACL) [Edit](#)

Grant basic read/write permissions to other AWS accounts. [Learn more](#)

This bucket has the bucket owner enforced setting applied for Object Ownership
When [bucket owner enforced](#) is applied, use bucket policies to control access. [Learn more](#)

Grantee	Objects	Bucket ACL
Bucket owner (your AWS account)		

Edit Object Ownership [Info](#)

Object Ownership
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☐ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

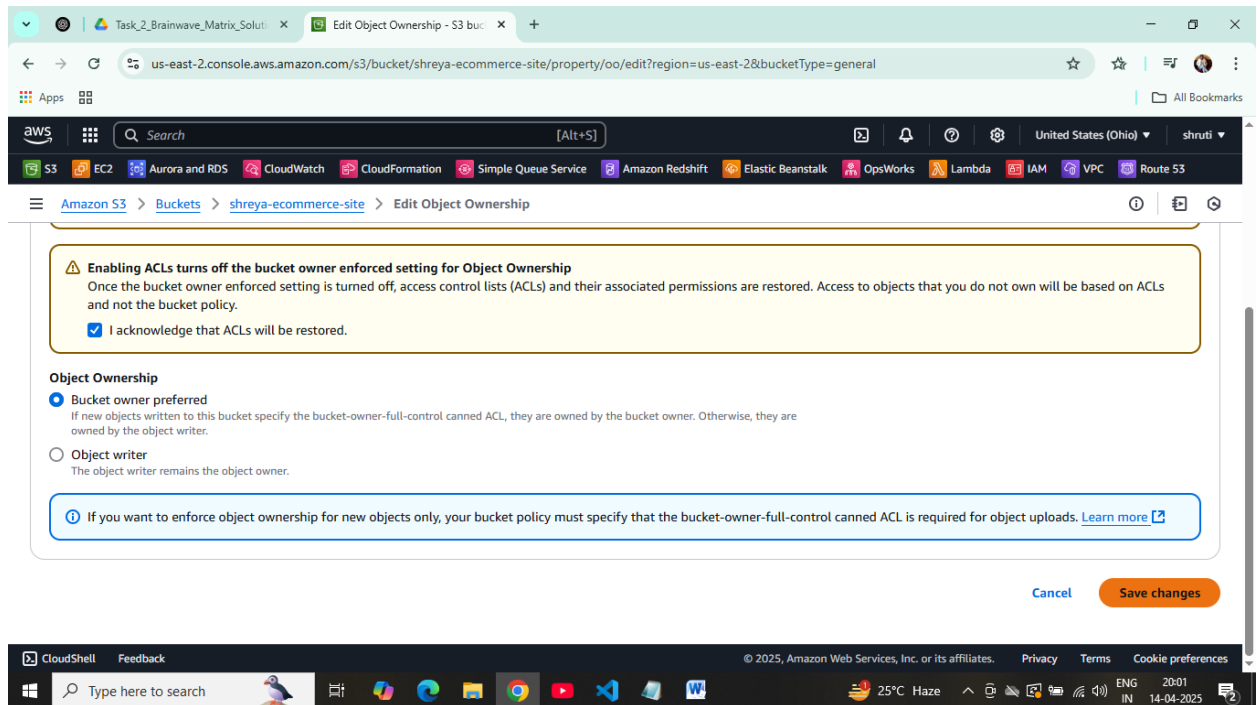
☒ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

⚠️ We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.

⚠️ Enabling ACLs turns off the bucket owner enforced setting for Object Ownership
Once the bucket owner enforced setting is turned off, access control lists (ACLs) and their associated permissions are restored. Access to objects that you do not own will be based on ACLs and not the bucket policy.

☐ I acknowledge that ACLs will be restored.

Object Ownership
☒ **Bucket owner preferred**
If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are



- Go to the bucket, click on object actions.
- Click on make public using ACL.
- Click on make public.
- Go to the bucket again, click on the object URL.
- The website will open.

The screenshot displays the AWS Management Console interface for an S3 bucket named 'shreya-ecommerce-site' in the 'us-east-2' region. The specific object being viewed is 'index.html'. The console provides a detailed overview of the object's properties, including its owner, region, last modified date, size, and type. Additionally, it lists the S3 URI, ARN, Etag, and Object URL. A context menu is open over the 'Object actions' button, showing a list of available actions such as 'Copy', 'Move', 'Initiate restore', 'Query with S3 Select', 'Edit actions', 'Rename object', 'Edit storage class', 'Edit server-side encryption', 'Edit metadata', 'Edit tags', and 'Make public using ACL'.

index.html Info

[Copy S3 URI](#) [Download](#) [Open](#) [Object actions](#)

Properties | **Permissions** | **Versions**

Object overview

Owner a60a674f61b73ed9801b7c48e6bee3750f4019553792d1c05f3172f576658220	S3 URI s3://shreya-ecommerce-site/index.html
AWS Region US East (Ohio) us-east-2	Amazon Resource Name (ARN) arn:aws:s3::shreya-ecommerce-site/index.html
Last modified April 14, 2025, 19:58:12 (UTC+05:30)	Entity tag (Etag) fc616998998f01a993991fef49459106
Size 2.2 KB	Object URL https://shreya-ecommerce-site.s3.us-east-2.amazonaws.com/index.html
Type	

Object actions

- Share with a presigned URL
- Calculate total size
- Copy
- Move
- Initiate restore
- Query with S3 Select
- Edit actions
 - Rename object
 - Edit storage class
 - Edit server-side encryption
 - Edit metadata
 - Edit tags
- Make public using ACL

Task_2_Brainwave_Matrix_Soluti

Make objects public - S3 bucke

us-east-2.console.aws.amazon.com/s3/buckets/shreya-ecommerce-site/object/edit_public_read_access?region=us-east-2&bucketType=general&showversions...

Apps

aws

Search

[Alt+S]

United States (Ohio) | shruti

Amazon S3

Buckets

shreya-ecommerce-site

Make public

Make public

Info

The make public action enables public read access in the object access control list (ACL) settings. [Learn more](#)

⚠ When public read access is enabled and not blocked by Block Public Access settings, anyone in the world can access the specified objects.

Specified objects

Find objects by name

Name	Type	Last modified	Size
index.html	html	April 14, 2025, 19:58:12 (UTC+05:30)	2.2 KB

Cancel

Make public

CloudShell

Feedback

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Task_2_Brainwave_Matrix_Soluti

index.html - Object in S3 bucke

us-east-2.console.aws.amazon.com/s3/object/shreya-ecommerce-site?region=us-east-2&bucketType=general&prefix=index.html

Apps

aws

Search

[Alt+S]

United States (Ohio) | shruti

Amazon S3

Buckets

shreya-ecommerce-site

index.html

index.html

Info

Copy S3 URI

Download

Open

Object actions

Properties

Permissions

Versions

Object overview

Owner

a60a674f61b73ed9801b7c48e6bee3750f4019553792d1c05f3172f576658220

AWS Region

US East (Ohio) us-east-2

Last modified

April 14, 2025, 19:58:12 (UTC+05:30)

Size

2.2 KB

Type

S3 URI

[s3://shreya-ecommerce-site/index.html](#)

Amazon Resource Name (ARN)

[arn:aws:s3:::shreya-ecommerce-site/index.html](#)

Entity tag (Etag)

[fc616998998f01a993991fef49459106](#)

Object URL

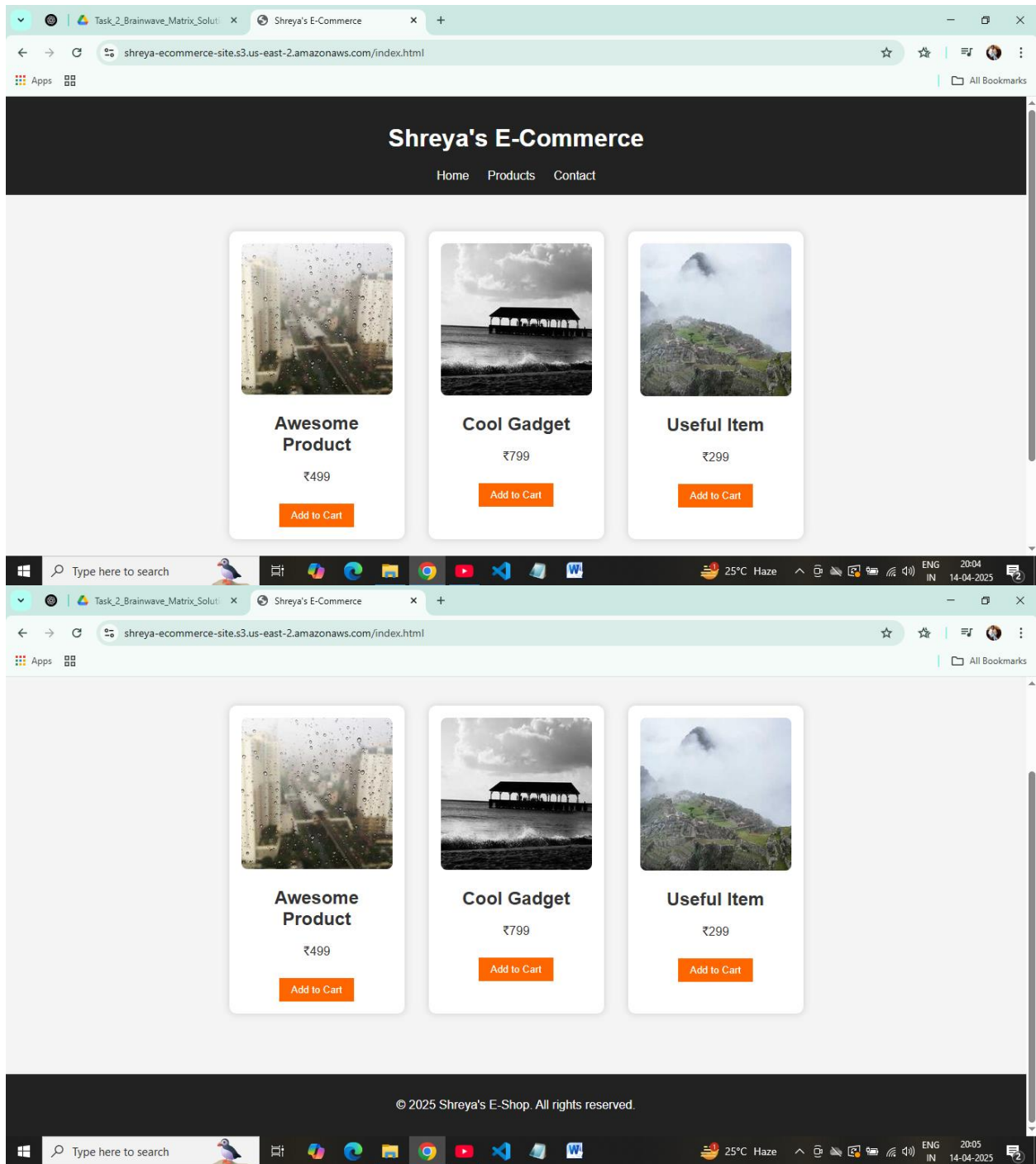
[https://shreya-ecommerce-site.s3.us-east-2.amazonaws.com/index.html](#)

https://shreya-ecommerce-site.s3.us-east-2.amazonaws.com/index.html

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Type here to search

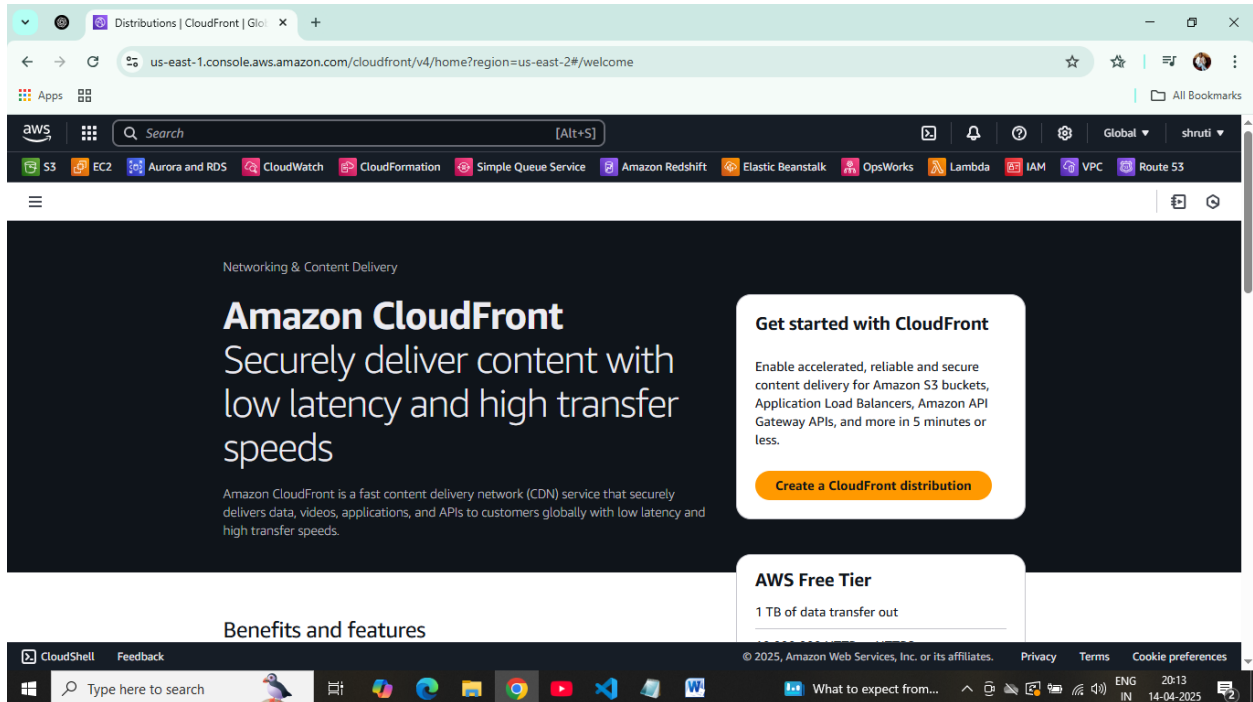
25°C Haze | 20:04 | 14-04-2025



OR

- We can access it using **CloudFront** as well.
- Go to CloudFront, click on create a **cloudfront distribution**.
- Give origin domain as the “s3 bucket static website endpoint”.

- Viewer protocol policy, select **HTTP and HTTPS**.
- Enable caching.
- Click on the **distribution domain name endpoint**, copy it.
- Paste in the browser, now again we can access our website.



Create distribution

Origin

Origin domain
Choose an AWS origin, or enter your origin's domain name. [Learn more](#)

shreya-e-commerce-site.s3-website-us-east-2.amazonaws.com

Enter a valid DNS domain name, such as an S3 bucket, HTTP server, or VPC origin ID.

Protocol [Info](#)

☒ HTTP only
☐ HTTPS only
☐ Match viewer

HTTP port
Enter your origin's HTTP port. The default is port 80.

80

Default cache behavior

Path pattern [Info](#)

Default (*)

Compress objects automatically [Info](#)

☐ No
☒ Yes

Viewer

Viewer protocol policy

☒ HTTP and HTTPS
☐ Redirect HTTP to HTTPS
☐ HTTPS only

Allowed HTTP methods

☒ GET, HEAD
☐ GET, HEAD, OPTIONS
☐ GET, HEAD, OPTIONS, PUT, POST, PATCH, DELETE

CloudFront

Distributions

Policies

Functions

Static IPs

VPC origins

What's new

▼ Telemetry

Monitoring

Alarms

Logs

▼ Reports & analytics

Cache statistics

Popular objects

Top referers

E2AEQB2XYU3MIZ

View metrics

General

Security

Origins

Behaviors

Error pages

Invalidation

Tags

Logging

Details

Distribution domain name

d381qa66pqps96.cloudfront.net

ARN

arn:aws:cloudfront::413240597866:distribution/E2AEQB2XYU3MIZ

Last modified

Deploying

Settings

Description

-

Alternate domain names

-

Standard logging

Off

Price class

Use all edge locations (best performance)

Cookie logging

Off

Edit

CloudShell

Feedback

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Type here to search

Gold -0.75%

ENG IN

20:18

14-04-2025

Distributions | CloudFront | Global

Shreya's E-Commerce

us-east-1.console.aws.amazon.com/cloudfront/v4/home?region=us-east-2#/distributions/E2AEQB2XYU3MIZ

All Bookmarks

Shreya's E-Commerce

Home Products Contact



Awesome Product

₹499

Add to Cart



Cool Gadget

₹799

Add to Cart



Useful Item

₹299

Add to Cart

Type here to search

Gold -0.75%

ENG IN

20:18

14-04-2025